Sequences, Progressions & Recurrences

RULE: TERM OR SUM? -> EXPLICIT OR RECURRENT? -> LOCAL RELATION OR GLOBAL FORMULA?
CAN NEARBY RELATIONS BE SUBTRACTED? -> CAN THE EXPRESSION TELESCOPE? -> ONLY THEN COMPUTE TERMS.

1. Identify the object

```
S_n=a_1+...+a_n => a_n=S_n-S_{n-1} (n>=2), a_1=S_1
```

Accumulated information and one-term information are different objects. If a sum is given but a term is requested, subtract adjacent totals.

2. AP, GP, or neither?

```
AP: a_{n+1}-a_n=d constant GP: a_{n+1}/a_n=r constant (nonzero terms)
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- 2,5,8,11,... -> AP. 2,6,18,54,... -> GP. 2,5,10,17,... -> neither; first differences are 3,5,7,...

Use the invariant to recognize the structure. Do not start by hunting for a memorized formula.

3. Explicit vs recurrent

- Explicit $a_n=f(n)$: the term is obtained directly from n.
- Recursive $a_{n+2}=F(a_{n+1},a_n)$: later terms depend on earlier terms; index range and enough initial values are part of the definition.
- To verify an explicit candidate: check initial values and substitute it into the recurrence on the full valid range.

Try before hints

If $S_n=3n^2-n$, find a_n and decide whether the resulting sequence is an AP. First attempt H0. Optional support only after attempting: H1 identify neighboring totals; H2 write S_n-S_{n-1}; H3 expand that difference.

Mandatory boundaries

term vs sum | AP vs GP vs neither | explicit vs recurrence | structural transform vs compute-many | algebraic recurrence vs counting-state recurrence

Local cancellation: windows, transforms, telescopes, invariants

Moving windows

```
W_i=a_i+...+a_{i+k-1} => W_{i+1}-W_i=a_{i+k}-a_i
```

All middle terms cancel. For fixed-length averages, multiply by the common length first. Increasing 4-term averages imply $a_{i+4}>a_i$; decreasing 7-term averages imply $a_{i+7}<a_i$. This is the mechanism behind IOQM-2025-Q26.

Transform a recurrence before iterating

```
a_{n+2}=3a_{n+1}-2a_n => a_{n+2}-a_{n+1}=2(a_{n+1}-a_n)
```

Thus first differences form a GP with ratio 2. A high index is a cue to compare nearby indices or test a target-shaped invariant before generating many raw terms.

Telescoping

```
1/[k(k+1)]=1/k-1/(k+1) => sum_{k=1}^n 1/[k(k+1)]=1-1/(n+1)
```

Consecutive factors often suggest rewriting each summand as $F(k)-F(k+1)$, so internal terms cancel and only boundary terms survive.

High-index neighboring-term invariant

For $a_{n+2}=p\, a_{n+1}+q\, a_n$, define

```
D_n=a_n^2-a_{n-1}a_{n+1}.
Direct substitution gives D_{n+1}=-q D_n.
```

For IOQM-2023-Q10: $a_0=1$, $a_1=-4$, $a_{n+2}=-4a_{n+1}-7a_n$. Then $D_{n+1}=7D_n$. Since $a_2=9$, $D_1=16-9=7$, hence $D_{50}=7^{50}$ and the number of positive divisors is 51.

Verification habit

When you invent an invariant, compute it at adjacent indices and prove its update law. A pattern seen in a few values is evidence, not proof.

> ADOPT: Compare neighboring indices so that most of the structure disappears.
> The same habit isolates a term, removes a moving window, simplifies a recurrence, and telescopes a sum.

Recognition + first-line lab and practice

Write only the first useful line or structural decision for 1-12; then solve the practice set. No method labels are supplied in the first attempt.

Recognition / first line

1. $S_n=2n^2+3n$; asked for a_n.
2. 5,9,13,17,...
3. 3,12,48,192,...
4. 5-term moving sums strictly increase.
5. $a_{n+2}=5a_{n+1}-4a_n$; study first differences.
6. sum $1/[k(k+1)]$.
7. Asked for a_{1000} from a second-order recurrence.
8. A recurrence is invented from counting tilings.
9. Verify a proposed explicit formula for a recurrence.
10. Every 4-term sum equals the preceding 4-term sum.
11. Denominator $(2k-1)(2k+1)$.
12. Target contains $a_n^2-a_{n-1}a_{n+1}$.

F0-F2 practice

13. Find the 15th term of 4,7,10,...
14. Find the 8th term of 3,6,12,...
15. If $S_n=n(n+1)$, find a_n.
16. Is 2,5,10,17,26,... AP, GP, or neither? Identify a simpler transformed pattern.
17. Every 4-term sum exceeds the preceding one. Relate a_i and a_{i+4}.
18. Evaluate $\sum_{k=1}^{20} 1/[k(k+1)]$.
19. If $S_n=2n^2-n$, show a_n is an AP and find its common difference.
20. Given $a_{n+2}=4a_{n+1}-3a_n$, show first differences form a GP.
21. Evaluate $\sum_{k=2}^{n} 1/[k(k-1)]$.
22. Let $a_1=1,a_2=4$, $a_{n+2}=3a_{n+1}-2a_n$. Find a closed form using first differences.

F3-F4 transfer and H0 mixed mastery

Changed-surface transfer

1. A sensor stores rolling totals of the last 6 readings; each new total is larger. Relate the entering and leaving readings.
2. All 3-term sums of a sequence are equal. Prove period dividing 3.
3. For $a_{n+2}=p a_{n+1}+q a_n$, prove $D_{n+1}=-qD_n$ for $D_n=a_n^2-a_{n-1}a_{n+1}$.
4. A machine recurrence is already given: list the ALG-04 checks before manipulating it.
5. A tiling problem requires you to invent the recurrence. Explain why defining the state and proving disjoint cases is combinatorial modelling, not generic sequence algebra.

H0 mixed mastery - no default hints

6. $S_n=4n^2+n$. Find a_n.
7. $a_{n+2}=5a_{n+1}-4a_n$, $a_1=2,a_2=5$. Find a_8 efficiently.
8. Evaluate $\sum_{k=1}^{50} 1/[k(k+1)]$.
9. Every 6-term sum exceeds the preceding one. State and prove the termwise inequality.
10. Every 4-term sum equals the preceding one. Prove period dividing 4.
11. Is 1,4,9,16,25,... AP or GP? Identify a simpler transformed sequence.
12. $a_1=3,a_2=7$, $a_{n+2}=3a_{n+1}-2a_n$. Find a_{20} without listing 20 terms.
13. Evaluate $\sum_{k=1}^{n} 1/[(2k-1)(2k+1)]$.
14. A path-count recurrence is derived from predecessor states. Separate the modelling step from later recurrence manipulation.
15. WHY NOT: a recurrence matches the first six values. Explain why this is not a proof.
16. WHY NOT: why is “AP or GP?” not a valid first question for every sequence problem?
17. State the clue that selected your first move for any two items above.

First-Step Reference + verified historical anchors

Recognition atlas

- Partial sums S_n, asked for a_n -> $a_n=S_n-S_{n-1}$.
- Constant first difference -> AP. Constant nonzero ratio -> GP. Otherwise do not force either label.
- Moving sums/averages -> subtract adjacent windows.
- High-index recurrence -> write nearby-index copies; test differences or a target-shaped invariant.
- Neighboring factors $k(k+1)$, $(k-1)k$, $(2k-1)(2k+1)$ -> try partial fractions / telescoping.
- Tilings/paths/states -> recurrence notation may be reused, but defining the counted state and deriving the recurrence belongs to combinatorial state modelling.

Quick recurrence verification

- Are enough initial values specified and correct?
- Does the recurrence hold over the stated index range?
- Does a proposed explicit formula satisfy both initialization and recurrence?
- If a transformed sequence/invariant is proposed, can its update law be proved independently?

Close contrast strip

term vs sum | AP vs GP vs neither | explicit vs recurrence | many-term iteration vs local transformation | moving-window cancellation vs telescoping | algebraic recurrence vs counting-state recurrence

Validated historical anchors

- IOQM-2025-Q26 - sliding-window averages -> adjacent-window subtraction. Independent answer verification: 10.
- IOQM-2023-Q10 - second-order recurrence -> neighboring-term determinant invariant. Independent answer verification: 51.

Exact historical wording remains under validated paper custody. Author-created practice above is not presented as historical IOQM material.

Source-independent checks used in this pack

- Q26: length 11 gives a strict directed cycle; a valid length-10 construction exists, so the maximum is 10.
- Q10: $D_{n+1}=7D_n$, $D_1=7$, so $D_{50}=7^{50}$ and its divisor count is 51.

ADOPT: Before generating terms, ask what changes locally and what cancels when neighboring indices are compared.
TRANSFER: the same structure appears in partial sums, rolling totals, recurrence transforms, high-index invariants and telescoping.